ODE-BASED APPROACH TO UNDERSTANDING RIOT DYNAMICS

ALICE LUNDGREN, EMMALINE MCKINNON, FLETCHER MURRAY, ASHLEY SPENCER

Abstract. We examine the growth of riots and protests using ODEs. We use a basic SIR model to model the spread of involvement, treating the growth of demonstrations as the spread of an epidemic. We incorporate elements of a predator-prey relationship to create a more robust model that accounts for the effects of retaliation against law enforcement. Analyzing the stability of equilibria, we evaluate the long-term behavior of incitements. In doing so, we gain a better understanding of the dynamics of real-world riots and how they can best be managed.

1. Background/Motivation

Mob mentality can transform charged situations into violent riots. Understanding how such events grow, spread, and decline can provide insights into social dynamics and inform strategies for law enforcement response and public safety. In this project, we aim to develop a mathematical framework to model the dynamics of crowd mobilization using ordinary differential equations (ODEs), SIR-type models, and predator–prey models.

Our focus is on events that include interactions between bystanders, violent protesters, and law enforcement. We aim to uncover relationships between the different groups, answering questions such as how the presence of violence or the reaction of law enforcement affect the growth of a riot.

Previous work, such as [BRS78] and [NMG14] have utilized SIR models to illustrate riot growth. Burbeck et al. note that the spread of rioting ideology is comparable to an epidemic; the more people currently "infected" and thus actively rioting, the faster the riot will grow. Additionally, if more people are closely concentrated the riot will spread faster. Therefore, both these papers treat bystanders, violent protesters, and ex-rioters as different 'compartments' with rioting spreading like an infection between the groups.

We build on this previous work in social movement modeling through the introduction of a law enforcement category. This compartment incorporates predator-prey-like terms to model the effect law enforcement has in containing the riot. Later, we incorporate recent criminology research from [Mag15] and [MBCL18] which describe a backlash effect to riot control. In essence, if law enforcement responds too aggressively or quickly, then there can be

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an increase in resentment, violence, and the rioting population. Through this inclusion we demonstrate that the latitude the public grants to the law enforcement's response greatly affects the riot dynamics.

2. Modeling

We note that for the entirety of the project all model variables are expressed as a fraction of the initial city population.

2.1. **Base Model.**

We begin with a standard SIR model:

$$\frac{dS}{dt} = -\alpha SP$$
$$\frac{dP}{dt} = \alpha SP - \beta P$$
$$\frac{dR}{dt} = \beta P$$

where

- S**:** susceptible individuals recruitable to the protest
- P**:** peaceful protesters
- R**:** individuals removed from protesting, whether voluntarily or involuntarily

Susceptible individuals join peaceful activism in proportion to activist numbers and eventually transition to the removed, non-participating category.

To this we add an additional category V of rioters who take violent action in response to the issue. We modify the susceptible and peaceful activist equations with $-\alpha_1 SV$ and $-\beta_1 PV$ to account for a small fraction being inspired to violence by current rioters. Additionally, at each time step ϕV violent activists transition to the removed category.

It follows from these modifications that we gain the following simple set of equations:

$$\frac{dS}{dt} = -\alpha_0 SP - \alpha_1 SV \qquad \frac{dR}{dt} = \beta_0 P + \phi V$$
$$\frac{dP}{dt} = \alpha_0 SP - \beta_0 P - \beta_1 PV \qquad \frac{dV}{dt} = \alpha_1 SV + \beta_1 PV - \phi V$$

Here we note that we initially enforce the condition $S+P+V+R=1$. We observe that this will remain constant for all time, since $\frac{dS}{dt}+\frac{dP}{dt}+\frac{dV}{dt}+\frac{dR}{dt} = 0$.

2.2. **Incorporating Law Enforcement.**

We incorporate law enforcement as a population of "predators" that attempt to suppress the population of violent rioters. We assume that that the number of arrests is proportional to the population of violent rioters and current officers. Furthermore, we assume that the rate at which law enforcement arrives on scene is proportional to the number of interactions with violent protesters, ie VL. Thus, as law enforcement has more interactions with violent protesters, more officers will be brought in to meet demand.

However, for ease of modeling we assume that the law enforcement population is separate from the protester population. Therefore, we do not stipulate that $S+P+V+R+L=1$ nor that $\frac{dS}{dt}+\frac{dP}{dt}+\frac{dV}{dt}+\frac{dR}{dt}+\frac{dL}{dt}=0$.

The above dynamics are captured with the following set of equations:

$$\begin{aligned}
\frac{dS}{dt} &= -\alpha_0 SP - \alpha_1 SV & \frac{dL}{dt} &= \varepsilon VL \\
\frac{dP}{dt} &= \alpha_0 SP - \beta_0 P - \beta_1 PV & \frac{dR}{dt} &= \beta_0 P + \phi VL \\
\frac{dV}{dt} &= \alpha_1 SV + \beta_1 PV - \phi VL
\end{aligned}$$

where

α_0**:** rate susceptible individuals become peaceful activists
α_1**:** rate susceptible individuals become violent activists
β_0**:** rate peaceful activists lose confidence in the cause
β_1**:** rate peaceful activists become violent
ϕ**:** rate of arrest per rioter interaction
ε**:** rate officers arrive per rioter interaction

2.3. **Benefits and Flaws of the Riot model.** Through experimentation, we find that the following set of parameters yield anticipated behavior:

$$\begin{aligned}
\alpha_0 &= 1 & \beta_0 &= 0.1 \\
\alpha_1 &= 0.1 & \beta_1 &= 1 \\
\phi &= 5 & \varepsilon &= 1
\end{aligned} \tag{1}$$

These parameters capture the assumptions that potential protesters quickly become peaceful but rarely turn violent. In contrast, hive-mind effects make peaceful protesters likely to become violent. In addition, officers efficiently remove violent protesters but are recruited slowly.

We thus generate Figure 1. The figure shows that as more people protest peacefully, the pool of potential protesters is reduced. After some delay, a violent contingent emerges, prompting more law enforcement to intervene. Eventually, the riot ends as participants are removed or lose interest. This matches the authors' intuitive understanding of riots; riots emerge quickly from peaceful demonstrations and are controlled by law enforcement.

However, such tame behavior is cherry-picked from the parameter space. Choosing the following yields Figure 2:

$$\begin{aligned}
\alpha_0 &= 1 & \beta_0 &= 0.1 \\
\alpha_1 &= 3 & \beta_1 &= 3 \\
\phi &= 1 & \varepsilon &= 1
\end{aligned}$$

These parameters assume officers efficiently remove violent protesters, officers arrive quickly to the scene, and violence occurs at a greater rate

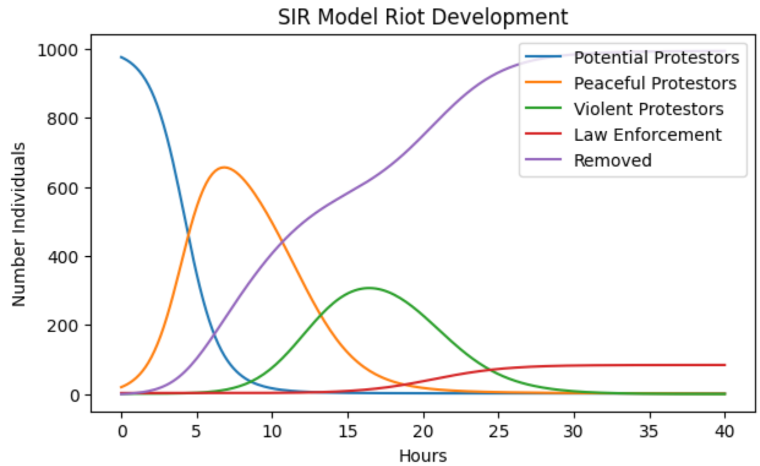

Figure 1. Evolution of protest intensity over time, showing increased violence and law enforcement response

than assumed by previous parameters. Here, an unreasonable amount of law enforcement, equal to the number of people in the city population, arrive to control the riot.

Furthermore, modern criminology research notes that there is a backlash effect to over-policing. For example in [MBCL18], researchers found that protesters' use of violence in the Occupy DC protests was "associated with perceptions of the extent to which police treat protesters in a procedurally unjust manner." This phenomenon is further established in [Mag15] where the National Commission on Civil Disorders determined that police overreaction in the 2014 Ferguson Missouri riots led to an increased resentment in the public and contributed to ongoing violence. Our model fails to capture this critical behavior and hence is unlikely to give accurate results.

2.4. **The Improved Model.** To address the issue of unreasonable police growth we introduce a limit $1 - \frac{L}{K_L}$ on the number of law enforcement in a certain city. Here K_L denotes the carrying capacity. In addition, we have added a $-\delta L$ term to capture the fact that a portion of officers are removed from the total policing population through events such as injury or the conclusion of a shift.

To model backlash from over-policing, we assume rioters become more violent once the policing population exceeds a threshold L_0. We incorporate this by multiplying the riot growth rate by a sigmoidal function of law enforcement. Few officers don't incite backlash but excessive policing increases the rate of violence. This is captured with the following term:

$$\frac{\rho}{1 + e^{-k(L-L_0)}} V \tag{2}$$

where

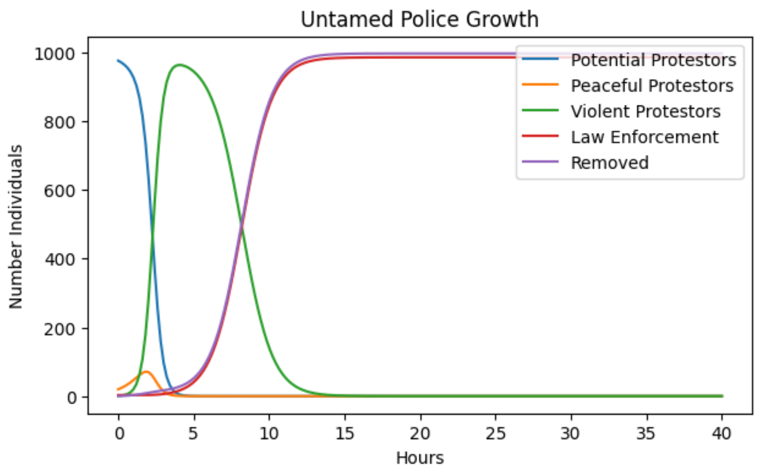

Figure 2. Unreasonable police growth with high violence and efficient policing parameters.

ρ: growth rate from backlash
k: sensitivity of the protest to policing
L_0: law enforcement threshold that results in backlash

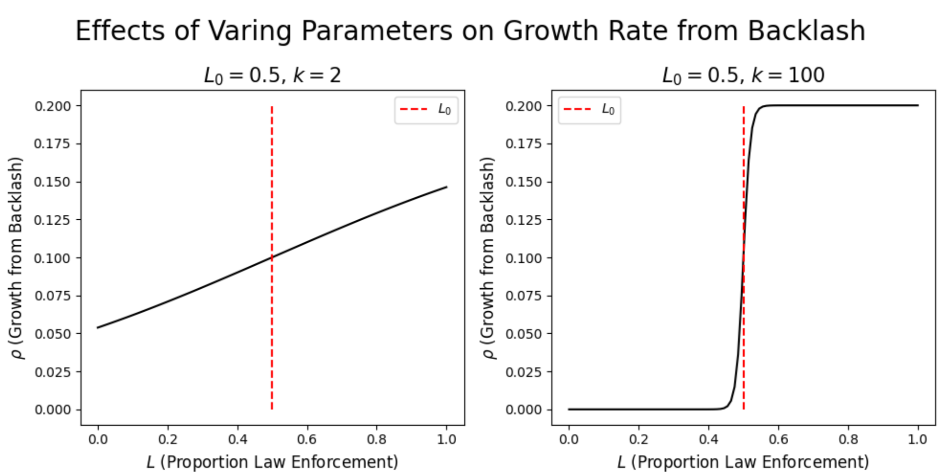

Figure 3. The effects of varying the parameters in 2. k controls the steepness of transition and hence the amount of law enforcement allowed in an area before backlash. L_0 describes the transition point where rioters start becoming more violent, and ρ is the max rate of growth.

Thus we create the following model:

$$
\begin{aligned}
\frac{dS}{dt} &= -\alpha_0 PS - \alpha_1 VS, \\
\frac{dP}{dt} &= \alpha_0 SP - \beta_0 P - \beta_1 PV, \\
\frac{dV}{dt} &= \left(\alpha_1 S + B_1 P + \frac{\rho}{1+e^{-k(L-L_0)}}\right) V - \phi VL, \\
\frac{dL}{dt} &= \varepsilon \left(1 - \frac{L}{K_L}\right) VL - \delta L, \\
\frac{dR}{dt} &= \beta_0 P + \phi VL
\end{aligned}
\tag{3}
$$

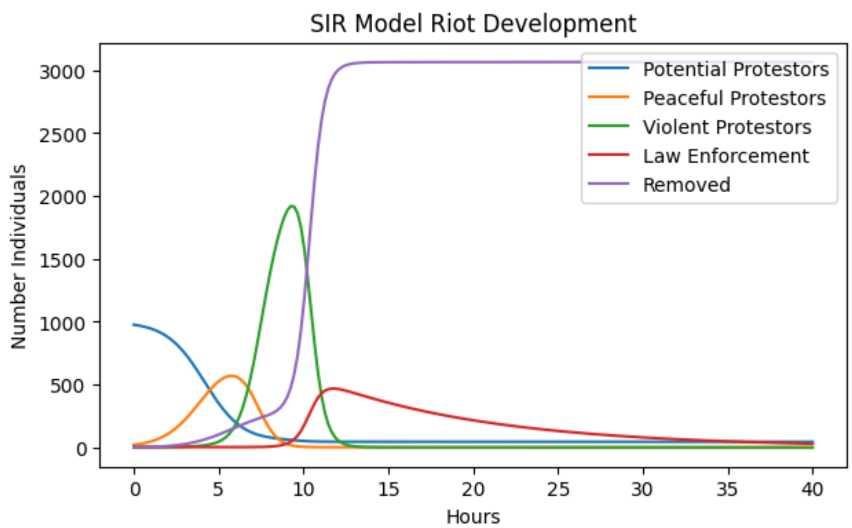

Figure 4. This figure illustrates the development of the riot using Equation 3 with the same parameters as 1 and adding the parameter of $k = 0.5$, $L_0 = 0.5$, and $\delta = 0.1$.

3. Results

Keeping the same parameters as 1 and adding the parameters of

$$
\begin{aligned}
k &= 0.5 \\
L_0 &= 0.5 \\
\delta &= 0.1
\end{aligned}
\tag{4}
$$

we generate Figure 4. Here we see better behavior for law enforcement where it does not surge to meet the total rioter population. Furthermore, we see a larger increase in the rioting population in response to law enforcement, demonstrating backlash. Thus, our new model remedies the issues of Figure 2 but retains the tame behavior of Figure 1.

Now we analyze the stability of our improved model by examining phase portraits and equilibria points. We focus our analysis on the behavior of rioters and law enforcement, as the question we seek to investigate is how these groups provoke each other. These rates—reiterated below—combine the impact of S and P on $\frac{dV}{dt}$ into a single term ρ_0.

$$\frac{dV}{dt} = \left(\rho_0 + \frac{\rho}{1 + e^{-k(L-L_0)}}\right) V - \phi V L$$
$$\frac{dL}{dt} = \varepsilon \left(1 - \frac{L}{K_L}\right) V L - \delta L$$

We begin by finding the equilibria points of $\frac{dV}{dt}$ and $\frac{dL}{dt}$, using the parameters from Equation 4 and 1. Setting V and L equal to 0 yields the obvious equilibrium (0, 0).

Due to the sigmoid function used in $\frac{dV}{dt}$, a non-zero equilibrium point cannot be found analytically. Newton's method gives a numerical estimation for L when $V \neq 0$. Setting $\frac{dV}{dt} = 0$ gives $\rho_0 + \frac{\rho}{1+e^{-k(L-L_0)}} = \phi L$. Iteratively running Newton's method, as shown in Equation 5, and solving $\frac{dL}{dt} = 0$ with the resulting L produces an estimate for the second equilibrium point: (0.0443, 0.1285).

$$L_{new} = L - \frac{\rho_0 + \frac{\rho}{1+e^{-k(L-L_0)}} - \phi L}{\phi + e^{-k(L-L_0)}(\phi - \phi L k + \rho_0 k)} \tag{5}$$

In order to investigate stability at equilibria points, we linearize around $\frac{dV}{dt}$ and $\frac{dL}{dt}$. The Jacobian is given by

$$DJ(L,V) = \begin{bmatrix} \left(\rho_0 + \frac{\rho}{1+e^{-k(L-L_0)}}\right) V - \phi L & \frac{\rho V k e^{-k(L-L_0)}}{(1+e^{-k(L-L_0)})^2} - \phi V \\ \varepsilon L(1 - \frac{L}{K_L}) & \varepsilon V(1 - 2\frac{L}{K_L}) - \delta \end{bmatrix}$$

To examine the stability of (0, 0), we substitute $V = L = 0$ into our Jacobian, which becomes

$$\begin{bmatrix} \rho_0 + \frac{\rho}{2} & 0 \\ 0 & -\delta \end{bmatrix}$$

with eigenvalues $\rho_0 + \frac{\rho}{2}$ and $-\delta$. Since ρ_0, ρ, and δ are positive, the eigenvalues have opposite signs, and thus $(0,0)$ is a saddle point.

Evaluating the Jacobian at $(0.0443, 0.1285)$, we find eigenvalues of -2.166 and -0.039. The negative eigenvalues reveal that this point is asymptotically stable. The phase portrait and equilibria points are shown in Figure 5. In this situation the riot reaches a steady-state of a permanent population rather than the desired extinction.

We will now examine a slight variation of the parameters. We work to discover a phase portrait where the number of violent protesters dies out quickly, instead of reaching a steady state. After experimentation, we find that k, the general public trust of law enforcement, affects this quality in the

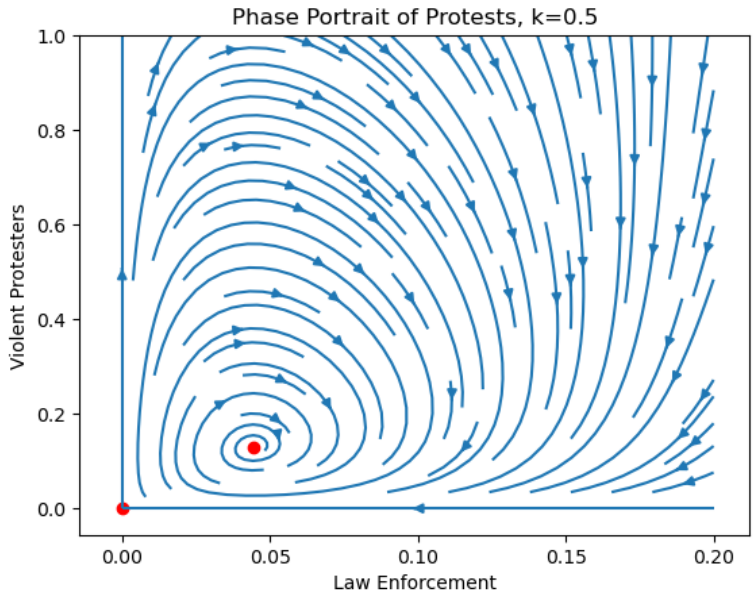

Figure 5. This figure represents the phase portraits of $\frac{dV}{dt}$ and $\frac{dL}{dt}$ with the parameters from Equation 4 and Equation 1. The equilibria points are shown in red.

phase portrait the most. Thus, we modify k to be 30, maintaining the other parameters defined in Figure 4. The phase portrait and equilibria points are shown in Figure 6.

4. Analysis/Conclusions

Through investigating the stability of equilibria across parameter values we find k is most influential in governing the lifecycle of a riot. When k is small, the system exhibits cyclical riot dynamics (see Fig. 5); however, when k is large, the riot eventually dies out (see Fig. 6). To interpret k we turn to Figure 3. The left panel (low k) illustrates a scenario where even small amounts of policing generates backlash, whereas the right panel (high k) represents a context in which substantial police action is required before backlash occurs. Thus, k can be interpreted as the amount of latitude the public or protesters grant to police intervention. As such, our model suggests to law enforcement that effective riot control depends on fostering conditions like public trust and perceived safety, which increase latitude.

However, our model exhibits significant flaws with room for improvement. Foremost among them is that we lack data with which to verify our hypotheses. Therefore, all conclusions are justified qualitatively based on the authors' limited understanding of the sociological theory of riots. Furthermore, our improved model has broken the assumption that the total population of

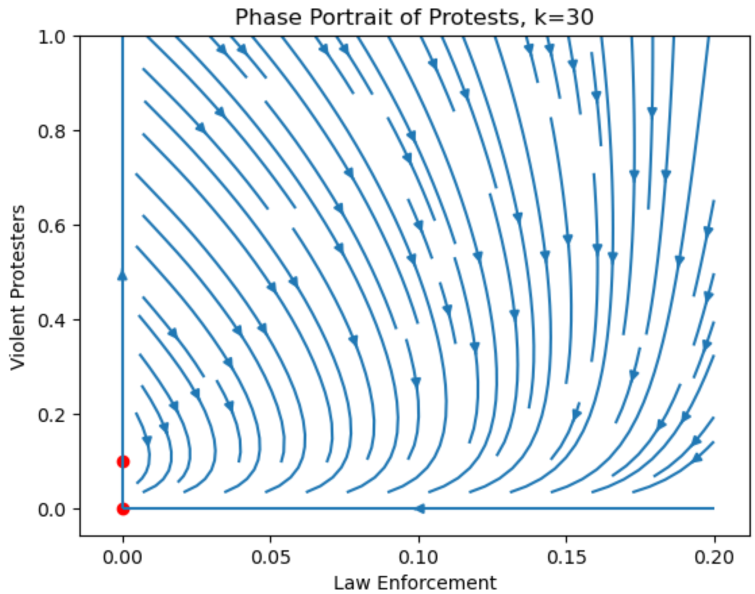

Figure 6. This figure represents the phase portraits of $\frac{dV}{dt}$ and $\frac{dL}{dt}$ with the parameters from Equation 4 and Equation 1 setting $k = 30$. The equilibria points are shown in red. Unlike the previous figure which produces cyclical dynamics, here we see near extinction of the riot population.

potential rioters must remain constant over time, (see Figure 4 where the rioter population exceeds the initial total amount of potential protesters). In future research, this issue may be remedied by scaling the backlash term in proportion to the population of potential and peaceful rioters and subtracting the correct proportion term from $\frac{dP}{dt}$ and $\frac{dS}{dt}$, respectively. Finally, our model exhibits an extreme number of parameters and constants, so future work in nondimensionalization can be done to reduce complexity.

Despite these shortcomings, our model provides a starting point for law enforcement and policymakers to consider when making decisions about riot control. Our novel approach of including the backlash effect of over-policing provides valuable insights into the relationship between rioters and police. Our findings—that law enforcement should increase public trust and safety in order to gain latitude for riot control—may help ensure a safer public and prevent unnecessary damage.

References

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